

FeralCTF — Technical Specification v1.0



Figure 1: FeralCTF logo

Purpose of this document: Complete implementation specification for the FeralCTF platform. Intended readers: human collaborators, AI coding agents (e.g. Qwen, OpenCode). This document is self-contained. Do not assume prior context. All decisions are documented here.

0. Project Summary

FeralCTF is a self-hosted Capture the Flag competition platform targeting non-profit and academic competitions that run on minimal infrastructure. The platform ships as a **single binary** that embeds frontend

assets and schema migrations and requires no external runtime dependencies (no Docker, no Node.js, no Python, no database server).

Primary constraints:

- Binary size: < 15 MB
- RAM at idle: < 50 MB
- RAM under load (200 concurrent users): < 200 MB
- CPU: runs on 1–2 vCPU comfortably
- Storage: < 10 MB database for a 500-team / 50-challenge competition

Feature target: Core CTFd-style workflows for small-to-medium competitions with a simpler single-binary operational model.

1. Technology Stack

1.1 Language & Runtime

- **Language:** Rust (stable toolchain, minimum 1.75)
- **Async runtime:** Tokio (multi-threaded, work-stealing)
- **Web framework:** Axum 0.8.x

No other runtime is required. The binary is fully self-contained.

1.2 Database

- **Engine:** SQLite via `rusqlite` + `r2d2_sqlite`
- **Mode:** WAL (Write-Ahead Logging) + `synchronous=NORMAL`
- **Connection pool:** `r2d2` pool of SQLite connections
- **Rationale:** CTF workloads are ~95% reads. WAL allows unlimited concurrent readers with a single non-blocking writer. SQLite is adequate; no Postgres/MySQL needed.
- **Backup:** SQLite backup API via `GET /api/admin/backup`; `sqlite3 ctf.db ".backup ..."` also works operationally.
- **Alternative backend:** None in the current implementation. SQLite is the supported storage engine.

1.3 Key Crates

Crate	Version	Purpose
axum	0.8.x	HTTP router, handlers, middleware
tokio	1.x	Async runtime
rusqlite	0.32.x	SQLite queries, migrations, backup
r2d2 / r2d2_sqlite	0.8.x / 0.25.x	SQLite connection pooling
rust-embed	8.x	Embed frontend assets into binary at build time
argon2	0.5.x	Password hashing (Argon2id)
jsonwebtoken	10.x	JWT auth (HS256)
serde / serde_json	1.x	JSON serialization, import/export
tower-http	0.6.x	CORS, compression, security-related middleware
tracing	0.1.x	Structured JSON logging
zip	8.x	Challenge file attachment packaging
dashmap	6.x	Supporting concurrent maps where needed

1.4 Frontend

- **Stack:** Vanilla JavaScript SPA + custom CSS. No React, Vue, or build pipeline.

- **Delivery:** Embedded into the binary via `rust-embed` at compile time.
- **Bundle size:** < 200 KB uncompressed, < 60 KB gzipped
- **Real-time:** WebSocket for live scoreboard events; clients may fall back to polling `/api/scoreboard`
- **Theme:** Dark terminal aesthetic (monospace throughout), light mode toggle available
- **Responsive:** Yes — mobile-friendly for on-site competitions

2. Architecture

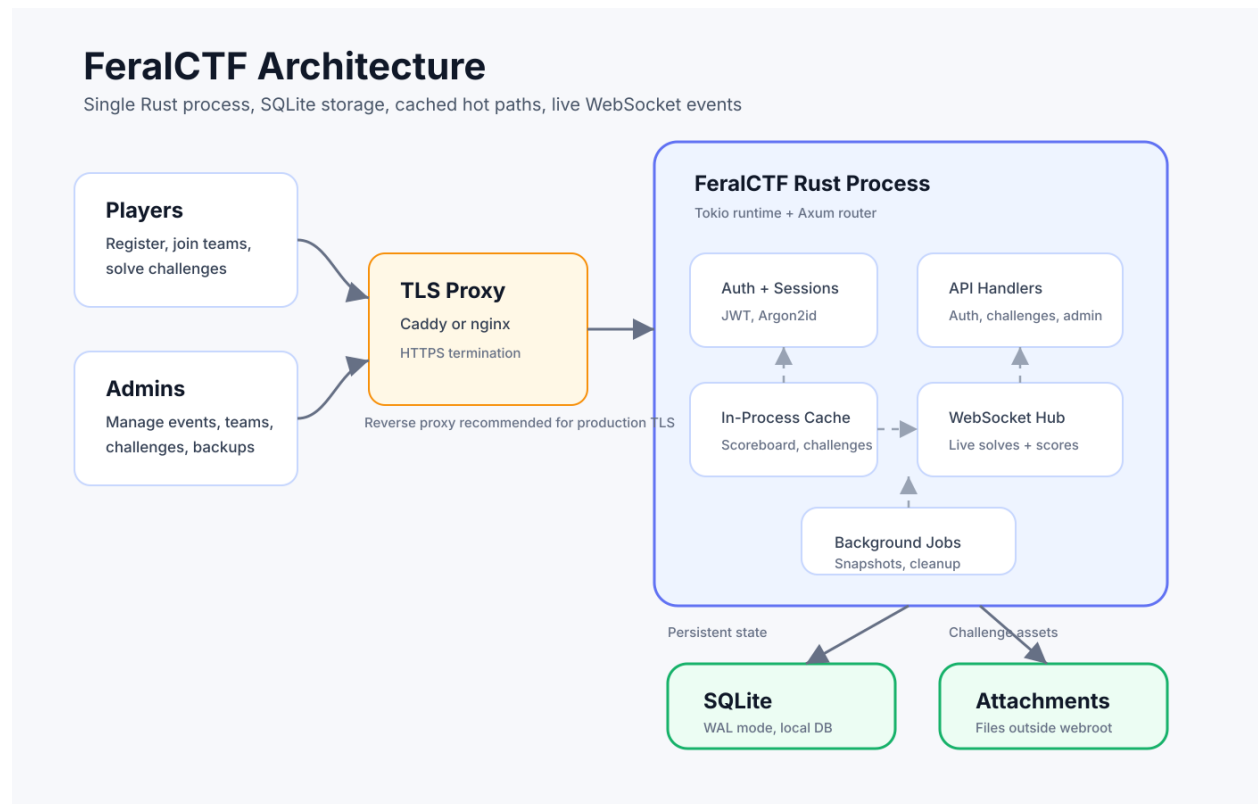


Figure 2: FeralCTF system architecture

2.1 Process Model

FeralCTF runs as a **single OS process**. All subsystems run inside one Tokio runtime:

```
feralctf [--config config.toml] [--port 8080]
```

On startup the binary:

1. Runs database migrations (idempotent)
2. Loads `config.toml` (or environment variable overrides)
3. Starts Axum HTTP server on configured port
4. Starts background workers:
 - Score history snapshot task
 - Submission rate limiter GC (every 60s)
 - WebSocket broadcast hub

2.2 Caching

All hot-path reads are served from in-process memory. The database is only hit on writes and cache misses.

Cache	Type	Invalidation
Scoreboard	<code>RwLock<Option<ScoreboardState>></code>	On accepted flag submission and score-affecting admin/team actions
Challenge list	<code>RwLock<Option<Vec<Challenge>>></code>	On admin challenge create/update/delete
User sessions	SQLite <code>sessions</code> table	TTL-based with server-side revocation
Rate limit counters	In-memory <code>RateLimiter</code> state	GC every 60s

2.3 Request Latency Targets

Layer	Component	Target
TLS	Reverse proxy (nginx/Caddy) — not built-in	< 1 ms
Rate limiting	In-process team/challenge submission limiter	< 0.1 ms
Auth	JWT validation, cache lookup	< 0.5 ms
Handler	Business logic	< 1 ms (cached)
Cache	in-process lock read	< 0.1 ms
Database	r2d2 SQLite pool	< 5 ms

2.4 WebSocket Architecture

A single `tokio::sync::broadcast` channel serves all connected WebSocket clients.

Event types:

```
enum WsEvent {  
    NewSolve    { team: String, challenge: String, points: u32, first_blood: bool },  
    Announcement { title: String, body: String },  
    StateChange { started: bool, ended: bool, frozen: bool },  
    ScoreUpdate  { scoreboard: Vec<TeamScore>, total_visible_points: i64 },  
}
```

- Max tested: 500 concurrent WS connections on 1 vCPU
- Fallback chain: WebSocket → 30s polling (no SSE; clients poll `/api/scoreboard`)

3. Database Schema

All tables use `INTEGER PRIMARY KEY` (SQLite rowid alias). Foreign key enforcement is at the application layer (not SQLite FK pragma) to avoid per-connection overhead.

3.1 Table Definitions

```
CREATE TABLE users (  
    id          INTEGER PRIMARY KEY,  
    username    TEXT NOT NULL UNIQUE,
```

```

    email            TEXT UNIQUE,
    password_hash    TEXT NOT NULL,           -- Argon2id
    role             TEXT NOT NULL DEFAULT 'player', -- admin / player / spectator
    team_id          INTEGER REFERENCES teams(id),
    created_at       INTEGER NOT NULL         -- Unix timestamp
);

CREATE TABLE teams (
    id                INTEGER PRIMARY KEY,
    name              TEXT NOT NULL UNIQUE,
    invite_code       TEXT NOT NULL UNIQUE,
    score             INTEGER NOT NULL DEFAULT 0,
    last_solve_at     INTEGER                -- Unix timestamp, for tiebreaking
);

CREATE TABLE challenges (
    id                INTEGER PRIMARY KEY,
    slug              TEXT NOT NULL UNIQUE,   -- stable identifier for import/export
    title             TEXT NOT NULL,
    description        TEXT NOT NULL,         -- Markdown, sanitized on render
    category           TEXT NOT NULL,
    flag_hash         TEXT NOT NULL,         -- sha256(lowercase(trim(flag)) + flag_salt)
    flag_salt         TEXT NOT NULL,
    flag_type         TEXT NOT NULL DEFAULT 'static', -- static / regex / dynamic
    flag_case_sensitive INTEGER NOT NULL DEFAULT 0,
    points            INTEGER NOT NULL,
    max_points        INTEGER NOT NULL DEFAULT 500,
    min_points        INTEGER NOT NULL DEFAULT 50,
    decay_rate        INTEGER NOT NULL DEFAULT 12,
    author            TEXT,
    tags              TEXT,                  -- JSON array string
    unlock_requires   INTEGER REFERENCES challenges(id),
    is_hidden         INTEGER NOT NULL DEFAULT 1, -- 0=visible, 1=hidden/draft, 2=archived
    created_at       INTEGER NOT NULL
);

CREATE TABLE solves (
    id                INTEGER PRIMARY KEY,
    team_id          INTEGER NOT NULL REFERENCES teams(id),
    user_id          INTEGER NOT NULL REFERENCES users(id),
    challenge_id     INTEGER NOT NULL REFERENCES challenges(id),
    solved_at        INTEGER NOT NULL,
    UNIQUE(team_id, challenge_id)
);

CREATE TABLE submissions (
    id                INTEGER PRIMARY KEY,
    team_id          INTEGER NOT NULL,
    user_id          INTEGER NOT NULL,
    challenge_id     INTEGER NOT NULL,
    flag             TEXT NOT NULL,         -- raw submitted value, for audit
    is_correct       INTEGER NOT NULL,
    ip_address       TEXT,
    submitted_at     INTEGER NOT NULL
);

```

```

);

CREATE TABLE hints (
    id            INTEGER PRIMARY KEY,
    challenge_id  INTEGER NOT NULL REFERENCES challenges(id),
    content       TEXT NOT NULL,
    cost_points   INTEGER NOT NULL DEFAULT 0,
    sort_order    INTEGER NOT NULL DEFAULT 0
);

CREATE TABLE hint_unlocks (
    id            INTEGER PRIMARY KEY,
    team_id       INTEGER NOT NULL,
    hint_id       INTEGER NOT NULL,
    points_deducted INTEGER NOT NULL,
    unlocked_at   INTEGER NOT NULL,
    UNIQUE(team_id, hint_id)
);

CREATE TABLE files (
    id            INTEGER PRIMARY KEY,
    challenge_id  INTEGER NOT NULL REFERENCES challenges(id),
    filename      TEXT NOT NULL,
    storage_path  TEXT NOT NULL,           -- path on disk, outside webroot
    size_bytes    INTEGER NOT NULL,
    sha256        TEXT NOT NULL
);

CREATE TABLE announcements (
    id            INTEGER PRIMARY KEY,
    title         TEXT NOT NULL,
    body          TEXT NOT NULL,
    challenge_id  INTEGER REFERENCES challenges(id),
    is_visible    INTEGER NOT NULL DEFAULT 1,
    created_at    INTEGER NOT NULL
);

CREATE TABLE score_history (
    id            INTEGER PRIMARY KEY,
    team_id       INTEGER NOT NULL,
    score         INTEGER NOT NULL,
    recorded_at   INTEGER NOT NULL       -- sampled every 5 min for graph
);

CREATE TABLE sessions (
    id            INTEGER PRIMARY KEY,
    user_id       INTEGER NOT NULL,
    token_hash    TEXT NOT NULL UNIQUE,  -- sha256 of JWT
    expires_at    INTEGER NOT NULL,
    revoked       INTEGER NOT NULL DEFAULT 0
);

CREATE INDEX idx_solves_team    ON solves(team_id);
CREATE INDEX idx_solves_chal    ON solves(challenge_id);

```

```
CREATE INDEX idx_submissions_team ON submissions(team_id, challenge_id);
CREATE INDEX idx_score_history ON score_history(team_id, recorded_at);
CREATE INDEX idx_sessions_token ON sessions(token_hash);
```

3.2 Dynamic Scoring Formula

$\text{points} = \max(\text{min_points}, \text{ceil}(\text{max_points} - \text{decay_rate} * (\text{solves} - 1)^2))$

Default parameters: max_points=500, min_points=50, decay_rate=12

This produces:

- 1st solve: 500 pts
- 2nd solve: 488 pts
- 3rd solve: 452 pts
- 6th solve: ~50 pts (clamped)

Static scoring (fixed points value, decay ignored) is the default for new challenges.

4. REST API

All endpoints return `application/json`. Authentication uses `Authorization: Bearer <JWT>`. Submission rate limits are enforced per team and per challenge (see §6.5).

4.1 Auth

Method	Path	Auth	Description
POST	<code>/api/auth/register</code>	None	Register user; first user becomes admin
POST	<code>/api/auth/login</code>	None	Returns signed JWT
POST	<code>/api/auth/logout</code>	Required	Revokes token in <code>sessions</code> table
GET	<code>/api/auth/me</code>	Required	Current user + team info
PUT	<code>/api/auth/password</code>	Required	Change password

Login response:

```
{
  "token": "<JWT>",
  "expires_at": 1762300800,
  "user": { "id": 1, "username": "n0tf0und", "role": "admin", "team_id": 3 }
}
```

4.2 Challenges

Method	Path	Auth	Description
GET	<code>/api/challenges</code>	Required	List visible challenges + per-team solve status
GET	<code>/api/challenges/:id</code>	Required	Challenge detail, hints (unlocked content), files
POST	<code>/api/challenges/:id/submit</code>	Required	Submit flag; rate limited 10/min/team

Method	Path	Auth	Description
POST	/api/challenges/:id/hint	Required	Unlock hint, deduct points
GET	/api/admin/challenges	Admin	List all challenges including hidden/draft
POST	/api/admin/challenges	Admin	Create challenge
PUT	/api/admin/challenges/:id	Admin	Update challenge (live during competition)
DELETE	/api/admin/challenges/:id	Admin	Delete challenge

Challenge list item:

```
{
  "id": 1,
  "slug": "jwt-jockey",
  "title": "JWT Jockey",
  "category": "web",
  "points": 100,
  "solve_count": 42,
  "solved_by_team": true,
  "is_hidden": false,
  "tags": ["jwt", "auth"],
  "file_count": 1,
  "hint_count": 2,
  "unlock_requires": null
}
```

Flag submission:

```
// Request
{ "flag": "flag{alg_none_ftw}" }

// Response (correct)
{ "correct": true, "points_earned": 100, "first_blood": false, "new_score": 2850 }

// Response (wrong)
{ "correct": false, "message": "Incorrect flag." }

// Response (rate limited)
{ "error": "rate_limited", "retry_after_seconds": 45 }
```

4.3 Scoreboard & Teams

Method	Path	Auth	Description
GET	/api/scoreboard	Optional	Full scoreboard (served from cache)
GET	/api/scoreboard/graph	Optional	Score-over-time for all teams (5 min samples)
GET	/api/teams/:id	Optional	Team profile + solve history
POST	/api/teams	Required	Create team
POST	/api/teams/join	Required	Join team by invite code
GET	/api/admin/teams	Admin	All teams with full stats

Method	Path	Auth	Description
POST	/api/admin/teams/:id/disqualify	Admin	Zero score, hide from scoreboard

Scoreboard response:

```
{
  "teams": [
    {
      "rank": 1,
      "team_id": 7,
      "team_name": "Example Team",
      "score": 100,
      "solve_count": 1,
      "last_solve_at": 1710000000
    }
  ],
  "total_visible_points": 1000,
  "generated_at": 1710000010
}
```

`total_visible_points` is the sum of current points for all visible challenges (`is_hidden = 0`) and is used by the frontend scoreboard progress bars. WebSocket `score_update` events include the same denominator so live updates preserve progress percentages.

4.4 Import / Export

Method	Path	Auth	Description
GET	/api/admin/export	Admin	Export full competition to JSON
GET	/api/admin/export?attachments=zip	Admin	JSON + companion attachments.zip
POST	/api/admin/import	Admin	Import from JSON (multipart/form-data)
POST	/api/admin/import?dry_run=true	Admin	Validate without writing

4.5 Admin Operations

Method	Path	Auth	Description
GET	/api/admin/submissions	Admin	Full submission log (paginated, filterable)
POST	/api/admin/announce	Admin	Broadcast announcement
GET	/api/admin/backup	Admin	Download raw SQLite database file
POST	/api/admin/competition/start	Admin	Start competition
POST	/api/admin/competition/end	Admin	End competition
POST	/api/admin/competition/freeze	Admin	Freeze scoreboard

5. Import / Export Format

5.1 Export

GET /api/admin/export returns a JSON file download. GET /api/admin/export?attachments=inline base64-encodes files < 5 MB inline. GET /api/admin/export?attachments=zip returns JSON only; attachments in separate ZIP.

Export schema:

```
{
  "feralctf_export_version": 1,
  "exported_at": "2026-11-08T18:00:00Z",
  "competition": {
    "name": "FeralCTF 2026",
    "dynamic_scoring": true,
    "score_freeze_minutes_before_end": 30,
    "max_team_size": 4
  },
  "categories": ["web", "crypto", "pwn", "forensics", "rev", "misc"],
  "challenges": [
    {
      "slug": "jwt-jockey",
      "title": "JWT Jockey",
      "category": "web",
      "description": "The admin forgot to validate the algorithm field.",
      "flag": "flag{alg_none_ftw}",
      "flag_type": "static",
      "flag_case_sensitive": false,
      "points": 100,
      "max_points": 500,
      "min_points": 50,
      "decay_rate": 12,
      "author": "n0tf0und",
      "tags": ["jwt", "auth"],
      "hints": [
        { "order": 1, "cost": 25, "content": "Think algorithm confusion attacks." },
        { "order": 2, "cost": 50, "content": "Try alg: none." }
      ],
      "files": [
        {
          "filename": "jwt_jockey.zip",
          "sha256": "a3f9c1...",
          "size_bytes": 14823,
          "data": "<base64>"
        }
      ],
      "unlock_requires": null,
      "is_hidden": false
    }
  ]
}
```

Security note: Flags are exported in plaintext. This endpoint is admin-only. Treat pre-competition exports as sensitive material.

What is included / excluded:

Field	Included	Notes
Competition metadata	Yes	Name, timing config, scoring config
All challenges	Yes	Including hidden/draft
Plaintext flags	Yes	Admin only — sensitive
Hints + costs	Yes	Full content and point costs
File attachments	Optional	Inline base64 or separate ZIP
Challenge tags	Yes	
Unlock dependencies	Yes	
Solve counts / submissions	No	Competition data stays in DB
User accounts / team scores	No	Export is challenge content only

5.2 Import

POST /api/admin/import accepts multipart/form-data:

Field	Type	Required	Description
file	JSON file	Yes	FeralCTF or CTFd export
attachments	ZIP file	No	Challenge attachment files
overwrite	bool	No (default: false)	Overwrite existing slugs
dry_run	bool	No (default: false)	Validate only, no DB writes

Dry run response:

```
{
  "valid": true,
  "challenges_to_create": 14,
  "challenges_to_skip": 0,
  "challenges_to_overwrite": 0,
  "attachment_warnings": [],
  "validation_errors": [],
  "preview": [
    { "slug": "jwt-jockey", "action": "create" },
    { "slug": "baby-rsa", "action": "create" }
  ]
}
```

Conflict resolution:

Scenario	Default (overwrite=false)	overwrite=true
Slug not in DB	Create	Create
Slug exists, identical	Skip (no-op)	Skip (no-op)
Slug exists, differs	Skip + warn	Overwrite
Attachment missing from ZIP	Import challenge, warn	Same
Invalid JSON	Reject entire import, no partial writes	Same
Unknown JSON field	Ignored (forward compat)	Same
CTFd format detected	Auto-convert via compat layer	Same

5.3 CTFd Compatibility

When a CTFd export ZIP is uploaded, the format is detected automatically.

CTFd Field	FeralCTF Field	Notes
name	title	Direct map
category	category	Direct map
description	description	HTML stripped to Markdown
value	points + max_points	Same value used for both
type: standard	flag_type: static	Warning flag set — review decay params
type: regex	flag_type: regex	
type: dynamic	flag_type: dynamic	
flags[].content	flag	First flag used; alternates stored
hints	hints	Cost + content mapped directly
files	files	Extracted from CTFd ZIP structure
tags	tags	Direct map

CTFd dynamic challenges get a warning flag on import because CTFd's decay formula differs from FeralCTF's. These challenges are imported as hidden and require admin review.

5.4 CLI Import

```
feralctf import challenges.json
feralctf import challenges.json --attachments ./files/ --overwrite
feralctf import challenges.json --dry-run
```

Recommended workflow: `--dry-run` first, review output, then import.

6. Security

6.1 Flag Storage

Flags are never stored in plaintext.

```
flag_hash = sha256(lowercase(trim(flag)) + challenge.flag_salt)
```

Each challenge has a unique random salt. Submission comparison hashes the user input and compares. A database dump cannot recover flags.

6.2 Password Storage

Argon2id with parameters: `memory=65536` (64 MB), `iterations=3`, `parallelism=2`. Target: ~300 ms per hash on the deployment target. Tune `memory` up/down to hit this target.

6.3 Authentication

- JWT HS256 signed with a server-generated secret (stored in config, not DB)
- Server-side revocation via `sessions` table (`token_hash` column)
- Default TTL: 24 hours for players, 4 hours for admin tokens
- Admin routes require an admin-role JWT and a valid non-revoked session.
- Admin management actions are logged with `user_id`, `action`, `target`, `detail`, `ip_address`, and `created_at`.

6.4 Input Validation

- All SQL uses `rusqlite` parameter binding (`params!` or equivalent).

- File attachments are stored outside the embedded frontend assets; import/export attachment ZIP handling is supported.
- Flag submissions: max 256 chars, stripped of leading/trailing whitespace
- Public challenge response types do not include flag hashes or salts.
- All API inputs validated via `serde` + custom validators before handler logic

6.5 Anti-Cheat

- Submission rate limit: 10 attempts/minute per team (configurable)
- Exponential backoff after 5 wrong attempts per challenge per team
- Flag sharing detection: same correct flag submitted by 2+ teams within configurable window → alert
- Submissions record IP address when available from request headers.
- Flag rotation: admin can update a challenge flag mid-competition; old hash invalidated immediately

6.6 HTTP Security Headers

Set by Tower middleware on all responses:

```
X-Content-Type-Options: nosniff
X-Frame-Options: DENY
Referrer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: default-src 'self'
```

The implemented CSP also allows WebSocket connections to same-origin `ws:/wss:` endpoints for live score-board updates.

7. Configuration

Full `config.toml` reference:

```
[server]
port = 8080
host = "0.0.0.0"
base_url = "https://ctf.yourdomain.com"
allowed_origins = []                                # optional CORS allowlist

[competition]
name = "FeralCTF 2026"
start_time = "2026-11-07T09:00:00-06:00"           # ISO 8601 with timezone
end_time   = "2026-11-08T17:00:00-06:00"
team_mode = true                                   # false = individual scoring
max_team_size = 4
registration_open = true
dynamic_scoring = true
score_freeze_minutes_before_end = 30               # 0 = no freeze

[database]
path = "./ctf.db"
backend = "sqlite"                                # or "json" for ephemeral

[auth]
jwt_secret = ""                                    # auto-generated on first run if empty
session_ttl_hours = 24
admin_session_ttl_hours = 4
```

```

[storage]
attachments_path = "./attachments"           # must be outside webroot
max_file_size_mb = 100

[rate_limit]
submissions_per_minute = 10
wrong_attempts_before_backoff = 5
backoff_base_seconds = 30
flag_sharing_window_seconds = 300

[notifications]
discord_webhook_url = ""                     # optional, for first blood + announcements

[logging]
level = "info"                               # trace / debug / info / warn / error
format = "json"                             # json / pretty

```

All config values can be overridden with environment variables using the prefix FERALCTF_:

```

FERALCTF_SERVER_PORT=9090
FERALCTF_SERVER_ALLOWED_ORIGINS=https://ctf.yourdomain.com
FERALCTF_DATABASE_PATH=/data/ctf.db
FERALCTF_AUTH_JWT_SECRET=supersecret

```

8. Deployment

8.1 Minimum Viable Deployment

```

# Download
curl -L https://github.com/yourorg/feralctf/releases/latest/download/feralctf-linux-x86_64 \
  -o feralctf && chmod +x feralctf

# Initialize (generates config.toml + empty DB)
./feralctf init

# Edit config.toml, then run
./feralctf --port 8080

# First user to register becomes admin

```

8.2 Reverse Proxy (nginx)

TLS is handled by the reverse proxy. FeralCTF handles WebSocket upgrade internally.

```

server {
    listen 443 ssl;
    server_name ctf.yourdomain.com;

    location / {
        proxy_pass http://127.0.0.1:8080;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}

```

```
}  
}
```

8.3 Systemd Service

```
[Unit]  
Description=FeralCTF Platform  
After=network.target  
  
[Service]  
Type=simple  
User=ctf  
WorkingDirectory=/opt/feralctf  
ExecStart=/opt/feralctf/feralctf  
Restart=on-failure  
MemoryMax=768M  
CPUQuota=150%  
  
[Install]  
WantedBy=multi-user.target
```

8.4 Backup

```
# Hot backup (no downtime required)  
sqlite3 ctf.db ".backup ctf-backup-$(date +%Y%m%d-%H%M).db"  
  
# Or via API  
curl -H "Authorization: Bearer $ADMIN_TOKEN" \  
https://ctf.yourdomain.com/api/admin/backup > backup.db
```

9. Build

9.1 Prerequisites

- Rust stable 1.75+
- cargo (bundled with Rust)
- Optional: cargo-watch for dev hot-reload

9.2 Commands

```
cargo run                                # dev build + run  
cargo build --release                    # optimized release (dynamically links libc)  
cargo run -- migrate                     # run DB migrations  
cargo run -- import challenges.json --dry-run # validate import bundle  
cargo test                              # run test suite  
cargo clippy --all-targets --all-features # lint all targets
```

Optional: fully static binary via musl The default release binary dynamically links against the host libc/libm/libgcc_s, which are present on all standard Linux distributions. For deployments where static linking is required (minimal container base images, airgapped environments), build against the musl target:

```
rustup target add x86_64-unknown-linux-musl  
cargo build --release --target x86_64-unknown-linux-musl
```

The resulting binary in `target/x86_64-unknown-linux-musl/release/feralctf` has no runtime shared-library dependencies.

9.3 Project Structure

```
feralctf/
src/
  main.rs          # CLI parsing, init/migrate/import, server startup
  config.rs        # Config struct, env var overrides, validation
  db/
    mod.rs         # connection pool setup, migration runner
  handlers/
    auth.rs        # register, login, logout, me, password
    challenges.rs   # list, detail, submit, hint unlock
    scoreboard.rs   # scoreboard (cache-served), graph data
    admin.rs       # challenge CRUD, user/team mgmt, competition controls
    ws.rs          # WebSocket hub, event broadcast
  models/
    user.rs        # User, Role
    team.rs        # Team
    challenge.rs    # Challenge, Hint, File, Submission, Solve
    scoreboard.rs   # TeamScore, ScoreboardState
  cache.rs         # in-process cache types and invalidation logic
  scoring.rs       # dynamic scoring formula, score recalculation
  anticheat.rs     # rate limiting, flag sharing detection, backoff
  storage.rs       # file storage helpers
  import_export.rs # JSON export, JSON import, CTFd compat adapter
  auth.rs          # JWT signing/verification, Argon2id hashing
  errors.rs        # unified error types, HTTP error responses
frontend/
  index.html       # SPA shell
  app.js           # all UI logic (challenges, scoreboard, admin, profile)
  style.css        # dark terminal theme
migrations/
  001_initial.sql
  002_audit_log.sql
Cargo.toml
README.md
```

10. Implementation Status

Current release: **1.0rc5**. All core sprints (0–13) complete. All items below verified against `cargo test` (50 tests passing).

Core

- ☒ SQLite schema + embedded migrations
- ☒ Config loading (file + environment overrides)
- ☒ Argon2id password hashing
- ☒ JWT auth with server-side session revocation
- ☒ User registration + login
- ☒ Team creation + invite-code join
- ☒ Challenge CRUD (admin)

- ☒ Challenge list endpoint with solve status per team (teamless users see challenges with `solved_by_team: false`)
- ☒ Flag submission (static)
- ☒ Flag submission (regex)
- ☒ Dynamic scoring formula
- ☒ Hint unlock with point deduction
- ☒ Scoreboard cache sorted by score and tiebreak
- ☒ Score history sampling background task
- ☒ WebSocket hub + event broadcast
- ☒ Submission rate limiting and backoff
- ☒ Frontend SPA (challenges, scoreboard, profile, admin)
- ☒ **rust-embed** frontend bundle

UI / UX

- ☒ Web-based user registration form (no out-of-band setup required)
- ☒ Admin navigation shown only to admin-role accounts
- ☒ Themed error page for unknown routes (404 and other HTTP error codes)
- ☒ Challenge cards — category color, difficulty dot, solve count, solved marker
- ☒ Category filter pills and title search on challenge grid
- ☒ Challenge visibility toggle (slider) in admin panel — player view syncs immediately
- ☒ Challenge edit modal in admin panel (title, category, points, description, flag, visibility)
- ☒ New challenges default to hidden; visibility opt-in at creation via toggle
- ☒ New challenges default to case-insensitive flag matching
- ☒ Challenge description renders URLs as clickable hyperlinks (open in new tab)
- ☒ Frontend layout aligned with specification mockups (§12)
- ☒ Brand icon in header

Import / Export

- ☒ JSON export endpoint (challenges + metadata)
- ☒ JSON export with inline base64 attachments
- ☒ JSON export with companion ZIP
- ☒ JSON import endpoint (dry-run mode)
- ☒ JSON import conflict resolution (skip / overwrite)
- ☒ CTFd export format detection + adapter
- ☒ CLI import subcommand (**feralctf import**)

Admin

- ☒ Challenge list endpoint returning all challenges including hidden (GET `/api/admin/challenges`)
- ☒ Competition start / end / freeze controls
- ☒ Announcement broadcast (WebSocket + stored)
- ☒ Submission log (paginated, filterable by team/challenge/result)
- ☒ User ban
- ☒ Team disqualify
- ☒ Challenge flag update through challenge update
- ☒ Database backup download endpoint

Security

- ☒ HTTP security headers
- ☒ Configurable CORS allowlist
- ☒ Flag storage as salted hash for static flags
- ☒ Admin action audit log

- ☒ IP logging on submissions when request headers provide an IP
- ☒ Flag sharing detection alert

Deployment

- ☒ `feralctf init` subcommand (generate config + empty DB + attachments directory)
- ☒ `feralctf migrate` subcommand
- ☒ Release build produces a self-contained binary with embedded frontend and migrations

Deferred or optional work remains for service packaging, CI/release automation, multi-architecture release artifacts, and advanced attachment validation.

11. Out of Scope (v1)

The following are explicitly deferred to keep v1 scope manageable:

- Per-challenge Docker spawner (netcat/pwn services) — separate optional module
- TOTP / 2FA — roadmap item
- Multi-instance replication (litestream / LiteFS) — roadmap item
- CTFtime.org API integration — roadmap item
- Writeup submission system — post-competition feature
- OAuth login (Discord, GitHub) — roadmap item

12. Glossary

Term	Definition
slug	URL-safe string identifier for a challenge, stable across import/export (e.g. <code>jwt-jockey</code>)
first blood	First team to solve a given challenge; may award bonus points
score freeze	Scoreboard stops updating N minutes before competition end; submissions still accepted
dynamic scoring	Challenge point value decreases as more teams solve it
WAL	Write-Ahead Logging — SQLite journal mode enabling concurrent reads
DQ	Disqualified — team score zeroed and hidden from public scoreboard
CTFd	Popular open-source CTF platform; FeralCTF supports importing its export format

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